# Repeat-purchase prediction: pipeline overview

Homework — Group 34 · what each stage does (left) and why (right)

### 1. Data

train.csv: 6,534 purchases (2015–2025), target 56% repeat
test.csv: 986 later purchases (2025–2026)
clientes.csv: optional customer table

**why**

Test lies after train in time, so validation must also predict later rows from earlier ones

### 2. Cleaning (fixed rules)

65 empty training rows dropped (none in test)
1e9 sentinels, negatives, max > total → missing
prior mean recomputed as total / count
1 prior purchase: days since = tenure

**why**

The same errors exist in test, so rows cannot be dropped: row-level rules learn nothing from the data and are applied to both files

### 3. Feature engineering

has_purchased_before, bought_30d, bought_90d
purchases_per_year, recent_share_30d / 90d
discount_rate (line + global / gross)

**why**

Recency and frequency drive repeat purchases (EDA); each feature uses one row, so no leakage

### 4. clientes.csv and excluded columns

2 regions + customer age joined and tested
customer_id: a label, not a measurement
purchase_date kept only to build the folds

**why**

clientes columns gave no measurable gain (age repeats tenure, rho 0.92) → not used; random_noise kept only as a probe

chosen on the folds

### 5. Preprocessing pipeline (fitted per fold)

numeric: log1p → IterativeImputer → scaling
0/1 flags: unchanged
categorical: 'missing' level → one-hot, levels with < 20 rows grouped

**why**

log1p: clearest effect (no log −0.006 AUC); iterative imputer better on 5/5 folds; scaling needed by the penalised model

### 6. Feature selection (votes)

filter: Spearman vs probe, redundancy > 0.9, chi-square for categoricals
embedded: L1 logistic regression (C = 0.1)
wrapper: RFECV — each run in all 5 folds

**why**

3 families, different blind spots; 15 votes per column, kept if ≥ 8 and above the probe (7) → 27 of 39 columns, same AUC as all 39

### 7. Model

logistic regression, C = 0.01
decision threshold 0.47
(from out-of-fold predictions)

**why**

C chosen first: largest effect (+0.016 AUC); threshold tuned for F1, the competition metric

### 8. Submission

refit on all training rows, predict test.csv
ID unchanged, sample_submission order
→ submission.csv

**why**

Same steps applied to test as to train; Kaggle public F1 0.780

## Assessment

5 time-ordered folds (TimeSeriesSplit) with a 90-day gap; every step fitted on the training part of each fold only. Options compared fold by fold: 'better' only beyond 2 standard errors and on all 5 folds. ROC-AUC to compare options, F1 on the repeat class for the final model.

**Results on the folds**

**F1 0.761 · ROC-AUC 0.79**

baselines (F1):
always 'repeat' 0.721
sample_submission rule 0.745